

Risk of Gestational Diabetes Mellitus in Relation to Maternal Egg and Cholesterol Intake

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Higher egg and cholesterol intakes are associated with increased risk of type 2 diabetes mellitus. However, their association with gestational diabetes mellitus (GDM) has not been evaluated. The authors assessed such associations in both a prospective cohort study (1996–2008; 3,158 participants) and a case-control study (1998–2002; 185 cases, 411 controls). A food frequency questionnaire was used to assess maternal diet. Multivariable models were used to derive relative risks and 95% confidence intervals. Compared with no egg consumption, adjusted relative risks for GDM were 0.94, 1.01, 1.12, 1.54, and 2.52 for consumption of ≤ 1 , 2–3, 4–6, 7–9, and ≥ 10 eggs/week, respectively (P for trend = 0.008). Women with high egg consumption (≥ 7 /week) had a 1.77-fold increased risk compared with women with lower consumption (95% confidence interval (CI): 1.19, 2.63). The relative risk for the highest quartile of cholesterol intake (≥ 294 mg/day) versus the lowest (< 151 mg/day) was 2.35 (95% CI: 1.35, 4.09). In the case-control study, the adjusted odds ratio for consuming ≥ 7 eggs/week versus < 7 eggs/week was 2.65 (95% CI: 1.48, 4.72), and the odds of GDM increased with increasing cholesterol intake (P for trend = 0.021). In conclusion, high egg and cholesterol intakes before and during pregnancy are associated with increased risk of GDM.